

```
In [3]: import pandas as pd

        from sklearn.model_selection import train_test_split
        from sklearn.preprocessing import StandardScaler
        from sklearn.neighbors import KNeighborsClassifier
        from sklearn.metrics import accuracy_score, confusion_matrix
```

```
In [5]: df = pd.read_csv("iris.csv")
```

```
In [13]: print(df.columns)
```

```
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
       'Species'],
      dtype='object')
```

```
In [14]: X = df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
        y = df['Species']
```

```
In [18]: X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random
```

```
In [19]: scaler = StandardScaler()

        X_train_scaled = scaler.fit_transform(X_train)
        X_test_scaled = scaler.transform(X_test)
```

```
In [28]: k = 5
        knn = KNeighborsClassifier(n_neighbors=k)
        knn.fit(X_train, y_train)
```

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Out[28]: ▼ KNeighborsClassifier ⓘ ?
```

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▶ Parameters
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In [29]: y_pred = knn.predict(X_test)
```

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In [30]: print("Accuracy:", accuracy_score(y_test, y_pred))

        print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

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Accuracy: 1.0
```

```
Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

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In [ ]:
```